

Lab 6
Jump and Loop Instructions
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Jump Instructions in 8086 Microprocessor

Following is the table showing the list of some jump instructions:

Instruction	Description	signed-ness	Flags
JO	Jump if overflow		OF = 1
JNO	Jump if not overflow		OF = 0
JS	Jump if sign		SF = 1
JNS	Jump if not sign		SF = 0
JE JZ	Jump if equal Jump if zero		ZF = 1
JNE JNZ	Jump if not equal Jump if not zero		ZF = 0
JB JNAE JC	Jump if below Jump if not above or equal Jump if carry	unsigned	CF = 1
JNB JAE JNC	Jump if not below Jump if above or equal Jump if not carry	unsigned	CF = 0
JBE JNA	Jump if below or equal Jump if not above	unsigned	CF = 1 or ZF = 1
JA JNBE	Jump if above Jump if not below or equal	unsigned	CF = 0 and ZF = 0

JL JNGE	Jump if less Jump if not greater or equal	signed	SF != OF
JGE JNL	Jump if greater or equal Jump if not less	signed	SF = OF
JLE JNG	Jump if less or equal Jump if not greater	signed	ZF = 1 or SF != OF
JG JNLE	Jump if greater Jump if not less or equal	signed	ZF = 0 and SF = OF
JP JPE	Jump if parity Jump if parity even		PF = 1
JNP JPO	Jump if not parity Jump if parity odd		PF = 0
JCXZ JECXZ	Jump if %CX register is 0 Jump if %ECX register is 0		%CX = 0 %ECX = 0

Processor Flags:

CF - carry flag - Set on high-order bit carry or borrow; cleared otherwise

PF - parity flag - Set if low-order eight bits of result contain an even number of "1" bits; cleared otherwise

ZF - zero flags - Set if result is zero; cleared otherwise

SF - sign flag - Set equal to high-order bit of result (0 if positive 1 if negative)

OF - overflow flag - Set if result is too large a positive number or too small a negative number (excluding sign bit) to fit in destination operand; cleared otherwise

LOOP Instructions of 8086 Microprocessor

The loop instructions cause the microprocessor to execute a series of instructions repeatedly. Basically, the LOOP instructions are short jump instructions on a condition i.e., when the condition satisfies a short jump is taken whose destination or target address is in the range of -128 bytes to +127 bytes from the instruction address after LOOP instruction. The various LOOP instructions and their functions are listed below.

Mnemonic	Meaning	Format	Operation
LOOP	Loop	LOOP short-label	Decrements CX by one, if CX $\neq$ 0, jump to "short-label".
LOOPE/LOOPZ	loop while equal/Loop while zero	LOOPE/LOOPZ short-label	Decrements CX by one, if CX $\neq$ 0 and ZF = 1, jump to "short-label".
LOOPNE/LOOPNZ	loop while not equal/Loop while not zero	LOOPNE/LOOPNZ short-label	Decrements CX by one, if CX $\neq$ 0 and ZF = 0, jump to "short-label".
JCXZ	Jump if CX is zero	JCXZ short-label	If CX = 0, branch to "short-label".

All the above instructions are iteration control instructions and are used to execute a series of instructions repeatedly on the successful satisfaction of the condition. The condition is to check the CX register and zero flag (ZF) or can be only to check the CX register. CX register must be loaded with the number of iterations to be performed, prior to entering the section of the code terminated by LOOP instructions. Let us see briefly each LOOP instruction.

Lab Work

- Take reference from the above instructions and implement following programs.
- Attach properly commented code in the lab report.
- Add flow chart of your logic for each program implemented.

Question 1:

Write a program that compares three numbers (num1=5, num2=8, num3=2) and stores maximum and minimum among them into max_num and min_num variable.

Question 2:

Write a program that takes a number (num1 = 5) and finds cumulative sum till that number and stores in my_sum variable. (5+4+3+2+1 = 15).

Note: Don't use $n*(n+1)/2$ formula. Use loops to generate the result.